Korea Technology and Information Promotion Agency for SMEs

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# Technology Proposal for 2026 TIPA* International Joint R&D Program

**(TIPA: Korea Technology and Information Promotion Agency for SMEs)**

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

## 1. Technology Proposer Information

| Proposer | Name | Pengjie Wang | Email | pengjiew@illinois.edu |
|---|---|---|---|---|
| Institution / Department | Physics, Materials Research Laboratory | | | |
| Address | 104 S Goodwin Ave, Urbana, IL 61801 | | | |

## 2. Technology Overview

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| Title of Proposed Technology | ***In-situ Synthesis of Two-Dimensional Contacts for 2D Semiconductors and Quantum Devices*** | | | |
| Korea's 12 National Strategic Technology Fields | Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum (√)<br>*(Please choose at least one of the areas above)* | | | |
| Industrial Technology Classification [1)] | **Technology Field 1** (Large Category) | | **Technology Field 2** (Large Category) | |
| | *Electrical・Electronics* | | | |
| | Medium Category | Small Category | Medium Category | Small Category |
| | *Semiconductor Devices and Systems* | *Other Semiconductor Devices* | | |
| Concept of Proposed Technology | This technology utilizes a "surface-confined two-dimensional mass transport" mechanism to synthesize high-quality metallic phases directly on monolayer materials [1]. A critical advantage of this approach is its low-temperature processing capability (typically occurring at temperatures significantly below bulk melting points). This "cold" metallization prevents thermal degradation of the delicate 2D lattice while enabling:<br>1. **Ohmic Contact Formation:** Creating atomically sharp, low-resistance interfaces that solve the Fermi level pinning problem in 2D semiconductors. [1]<br>2. **Induced Superconductivity for quantum applications:** Converting topological chalcogenides (e.g., $WTe_2$) into robust superconductors via controlled metallization (e.g., with Palladium). This enables the direct fabrication of superconducting-semiconducting hybrid structures essential for quantum logic. [2] | | | |

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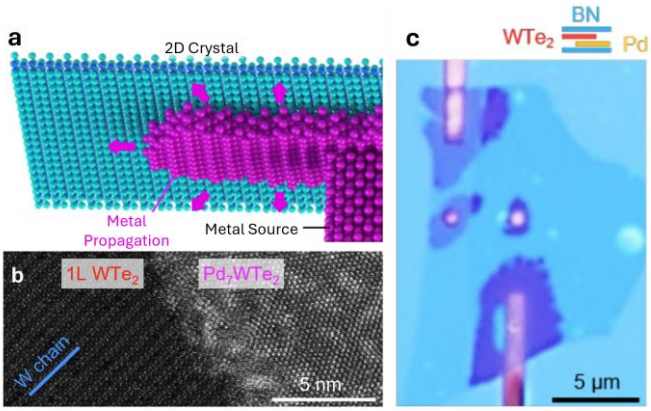
| |  <br>**Figure 1 \| 2D Metallization of monolayer $WTe_2$.** |
| **Rationale for Joint R&D Project** | The "contact problem" is the primary bottleneck for both 2D semiconductor electronics and topological quantum computing (3). Traditional physical vapor deposition is too energetic and hot, often damaging the 2D lattice. Our proposed solution offers a gentle, low-temperature, "chamber-free" synthesis pathway that creates pristine contacts and novel superconducting phases on demand. Collaboration with a Korean SME is critical to scale this process from a fundamental discovery to a reliable manufacturing technique for next-generation quantum devices and logic transistors. |
| **Keywords** | 1. 2D materials<br>2. Contact Resistance<br>3. Ohmic Contacts<br>4. Surface-Confined Metal Transport<br>5. Superconductivity |

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## 3. Research and Development Objectives & Plan

| Technology Readiness Level (TRL)[2] | Start of the Project | Level ( 2 ) | End of the Project | Level ( 5 ) |
|---|---|---|---|---|
| **R&D Objectives** | • **Material Exploration:** Systematically investigate and optimize the 2D metallization process using a diverse range of metals—specifically Nickel (Ni), Bismuth (Bi), Germanium (Ge), and Copper (Cu)—to identify the ideal contact materials for different 2D semiconductor hosts.<br>• **Process Optimization:** Establish protocols for low-temperature, surface-confined mass transport on key technological materials ($MoS_2$, $WSe_2$, $WTe_2$) to maximize diffusion length while preserving lattice integrity.<br>• **Quantum Device Engineering:** Leverage the metallization process to induce superconductivity in specific regions, creating monolithic Josephson junctions and hybrid quantum structures.<br>• **Interface Characterization:** Use tunneling spectroscopy to verify atomically sharp interfaces, the absence of Schottky barriers, and the proximity-induced superconducting gap. | | | |
| **R&D Target Specifications** | **Semiconductor Applications**<br>• **Contact Resistance:** < 100 Ω μm (approaching the quantum limit).<br>• **Scalability:** Uniform metallization over > 100 $\mu m^2$ areas.<br><br>**Quantum Applications:**<br>• **Superconducting Transition ($T_c$):** Demonstrate controlled superconducting transition temperatures ($T_c$) in metallized regions.<br>• **Phase Stability:** Validate stability of the metallized phases (e.g., $Pd_7WTe_2$) under cryogenic and operational conditions. | | | |
| **R&D Plan** | **Phase 1:**<br><br>**Material Discovery & Process Optimization (Months 1-6)**<br><br>• **Task 1.1:** Metal Screening: Deposit thin films of Ni, Bi, Ge, and Cu onto reference TMDs ($WSe_2$, $MoS_2$, $WTe_2$). Perform thermal annealing at varying temperatures (100°C - 400°C) to identify the onset of surface-confined mass transport.<br>• **Task 1.2: Kinetic Analysis:** Use atomic force microscopy (AFM) and Raman spectroscopy to measure diffusion lengths and reaction rates. Establish a "phase diagram" for each metal-semiconductor pair.<br>• **Milestone 1:** Selection of the top 2 metal candidates based on diffusion efficiency and phase stability.<br><br>**Interface Engineering & Characterization (Months 7-12)**<br><br>• **Task 2.1: Interface Spectroscopy:** Fabricate tunneling devices to probe the metal-semiconductor interface. Use low-temperature tunneling spectroscopy to screen for Schottky barriers and measure the superconducting gap in induced regions.<br>• **Task 2.2: Device Fabrication:** Design and fabricate Hall bar and FET structures using optimized metallization recipes.<br>• **Task 2.3: Benchmarking: Compare the electrical performance (mobility, on/off ratio, contact resistance) of "2D metallized" devices against standard evaporated contact devices.**<br>• **Milestone 2:** Demonstration of a 2D transistor with contact resistance superior to conventional Ti/Au contacts.<br><br>**Phase 2: Prototype Validation (Months 13+ / Future Phase)** | | | |

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| | • **Task 3.1:** Integration of metallized contacts into complex logic gates (inverters) and Josephson junctions (for quantum applications).<br>• **Task 3.2:** Reliability testing under bias stress and thermal cycling. |

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## 4. R&R (Roles and Responsibilities)

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| **Expected R&R of Research Institute** | • Lead the fundamental in-situ metallization synthesis and contact quality test measurements.<br>• Identify materials suitable for 2D contacts<br>• Characterize interfaces and interlayer structures.<br>• Characterized the induced superconducting phases and topological properties using advanced spectroscopy.<br>• Provide the initial "recipes" for surface-confined metallization. |
| **Expected R&R of Korean SME Partner** | • Evaluate the scalability of the low-temperature "on-chip" metallization process for wafer-scale manufacturing.<br>• Conduct reliability testing and integration of these high-quality contacts into commercial-relevant device architectures.<br>• Provide feedback on industrial compatibility (e.g., thermal budget, material constraints). |

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## 5. Expected Outcomes/Impact

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| **Expected Outcomes and Impact** | **Outcomes:**<br>**(Semiconductor)** A validated, non-damaging contact engineering technique that solves the Fermi level pinning issue in 2D semiconductors.<br>**(Quantum)** A high quality 2D superconductor will be realized. Key for fabrication of Josephson Junction, critical component for quantum computing.<br><br>**Impact:** This technology could be the critical technology for applying 2D material devices for transistor and quantum computation, enable the commercial viability of 2D material-based electronics by removing the largest parasitic resistance in the circuit. It supports the development of lower-power, higher-speed processors and novel quantum computing architectures. |

## 6. References

(1) Y. Jia et al., "Surface-confined two-dimensional mass transport and crystal growth on monolayer materials," Nat. Synth., vol. 3, no. 3, pp. 386-393, Mar. 2024.

(2) Y. Jia et al., "Superconductivity from On-Chip Metallization on 2D Topological Chalcogenides," Phys. Rev. X, vol. 14, 021051, 2024.

(3) "Contacts in 2D," Nat. Electron., vol. 5, no. 5, pp. 255-255, May 2022.

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## ☐ Terms:

### 1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (Appendix 1, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

### 2) Technology Readiness Level (TRL)

| Description | | |
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| ·A consistent and objective indicator of the technological maturity of key element technologies | | |
| | **Basic Research Stage (1-2)** | 1st Stage: Basic theories and experiments<br>2nd Stage: Proof of Concept, such as practical ideas or patents |
| | **Experimental Stage (3-4)** | 3rd Stage: Verification of basic performance at a laboratory scale<br>4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale |
| | **Prototype Stage (5-6)** | 5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems<br>6th Stage: Production and performance evaluation of prototypes at a pilot scale |
| | **Productization Stage (7-8)** | 7th Stage: Reliability testing and evaluation by potential customers<br>8th Stage: Prototype certification and standardization |
| | **Commercialization Stage (9)** | 9th Stage: Commercialization |

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